Final Project_Data Science

Tanner Rougeau

2026-10-25

Classmates Jason Spino and Lily Impicito consulted, as well as Claude for help in coding and logical framework for submission.

Load packages and data

```
# load tidyverse for data wrangling and ggplot2
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# read the covariate and gene expression data in its original state
covar <- read.csv("C:/Users/tanne/OneDrive/GradSchool/Foundations of Data Science/Project/GSE85307_covar
expr <- read.csv("C:/Users/tanne/OneDrive/GradSchool/Foundations of Data Science/Project/GSE85307_geneEx
# read the probe-to-gene-name linker file
genes <- read.csv("C:/Users/tanne/OneDrive/GradSchool/Foundations of Data Science/Project/GSE85307_genel
```

Inspect Covariance data

```
# show variable names, types, and first values in the covariate file
str(covar)
```

```
## 'data.frame':    157 obs. of  10 variables:
##  $ subjectID                  : chr  "6053887-M" "3064425-M" "6094898-M" "7050617-M" ...
##  $ geoID                      : chr  "GSM2264346" "GSM2264347" "GSM2264348" "GSM2264349" ...
##  $ mat_age_yrs                : chr  "age_years: 31.06639288" "age_years: 27.79466119" "age_years: 3
##  $ preg_preeclampsia          : chr  "pregnancy_condition: Preeclampsia" "pregnancy_condition: Norma
##  $ delivery_gestAge_wks       : chr  "gestation_weeks: 37.42857143" "gestation_weeks: 39.42857143"
```

```
##  $ sampleCollection_gestAge_wks: chr  "blood_draw_at_gestation_week: 10" "blood_draw_at_gestation_wee
##  $ batch_Microarray            : chr  "microarray_hybridization_date: 03/15/2013" "microarray_hybridi
##  $ mat_BMI                     : chr  "bmi_kg/m2: 37.52407538" "bmi_kg/m2: 20.59144273" "bmi_kg/m2: 1
##  $ mat_vitD                    : chr  "vitamind_25(0h)_ng/ml_whole_blood_baseline: 7.94" "vitamind_25
##  $ mat_asthma                  : chr  "asthma_0=healthy_1=asthmatic: 1" "asthma_0=healthy_1=asthmatic
```

```
# show summary statistics to spot missing codes or odd values
summary(covar)
```

```
##      subjectID         geoID        mat_age_yrs  preg_preeclampsia
##  Length   :157   Length   :157   Length   :157   Length   :157
##  N.unique :157   N.unique :157   N.unique :157   N.unique :  2
##  N.blank  :  0   N.blank  :  0   N.blank  :  0   N.blank  :  0
##  Min.nchar:  9   Min.nchar: 10   Min.nchar: 13   Min.nchar: 27
##  Max.nchar:  9   Max.nchar: 10   Max.nchar: 22   Max.nchar: 33
##  delivery_gestAge_wks sampleCollection_gestAge_wks  batch_Microarray
##  Length   :157        Length   :157                Length   :157
##  N.unique : 44        N.unique : 53                N.unique :  8
##  N.blank  :  0        N.blank  :  0                N.blank  :  0
##  Min.nchar: 19        Min.nchar: 32                Min.nchar: 41
##  Max.nchar: 28        Max.nchar: 41                Max.nchar: 41
##       mat_BMI         mat_vitD       mat_asthma
##  Length   :157   Length   :157   Length   :157
##  N.unique :140   N.unique :132   N.unique :  2
##  N.blank  :  0   N.blank  :  0   N.blank  :  0
##  Min.nchar: 20   Min.nchar: 45   Min.nchar: 31
##  Max.nchar: 22   Max.nchar: 48   Max.nchar: 31
```

Inspect gene expression data

```
# number of rows (probes) and columns (samples)
dim(expr)
```

```
## [1] 500 158
```

```
# first 5 column names to see how samples are labeled
names(expr)[1:5]
```

```
## [1] "X"          "GSM2264363" "GSM2264379" "GSM2264394" "GSM2264418"
```

Inspect gene name linker file

```
# structure of the probe-to-gene linker file
str(genes)
```

```
## 'data.frame':    750 obs. of  2 variables:
##  $ PROBEID: int  8128043 7962489 7920271 7900468 8026496 7989725 8102362 7930106 8035318 8041820 ...
##  $ GENE   : chr  "CNR1" "PLEKHA8P1" "S100A4" "NFYC" ...
```

Clean covariance data

```
# remove the label/text before each value in columns 3-10
covar[3:10] <- lapply(covar[3:10], function(x) sub(".*: ", "", x))

# convert the continuous covariates from text to numbers
covar$mat_age_yrs <- as.numeric(covar$mat_age_yrs)
covar$delivery_gestAge_wks <- as.numeric(covar$delivery_gestAge_wks)
covar$sampleCollection_gestAge_wks <- as.numeric(covar$sampleCollection_gestAge_wks)
covar$mat_BMI <- as.numeric(covar$mat_BMI)
covar$mat_vitD <- as.numeric(covar$mat_vitD)

# convert the categorical covariates to factors with readable labels
covar$preg_preeclampsia <- factor(covar$preg_preeclampsia, levels = c("Normal", "Preeclampsia"))
covar$mat_asthma <- factor(covar$mat_asthma, levels = c(0, 1), labels = c("Healthy", "Asthma"))
covar$batch_Microarray <- factor(covar$batch_Microarray)

# confirm after cleaning
summary(covar)
```

```
##      subjectID         geoID      mat_age_yrs       preg_preeclampsia
##  Length   :157   Length   :157   Min.   :18.43   Normal      :110
##  N.unique :157   N.unique :157   1st Qu.:22.15   Preeclampsia: 47
##  N.blank  :  0   N.blank  :  0   Median :26.57
##  Min.nchar:  9   Min.nchar: 10   Mean   :26.38
##  Max.nchar:  9   Max.nchar: 10   3rd Qu.:30.14
##                                  Max.   :38.76
##
##  delivery_gestAge_wks sampleCollection_gestAge_wks   batch_Microarray
##  Min.   :26.43        Min.   :10.00                03/19/2013:45
##  1st Qu.:38.14        1st Qu.:12.14                03/15/2013:44
##  Median :39.29        Median :13.43                03/22/2013:29
##  Mean   :38.82        Mean   :14.13                03/27/2013:26
##  3rd Qu.:40.00        3rd Qu.:16.43                04/04/2013:10
##  Max.   :42.00        Max.   :18.86                03/18/2013: 1
##                                                    (Other)   : 2
##     mat_BMI         mat_vitD       mat_asthma
##  Min.   :17.95   Min.   : 5.11   Healthy:94
##  1st Qu.:22.90   1st Qu.:13.40   Asthma :63
##  Median :27.33   Median :20.40
##  Mean   :28.94   Mean   :22.98
##  3rd Qu.:33.27   3rd Qu.:29.50
##  Max.   :61.95   Max.   :80.80
##
```

Verify sample matching

```
# check that expression sample columns have  matching geoID in the covariate data
all(names(expr)[-1] %in% covar$geoID)
```

```
## [1] TRUE
```

```r
# check that covariate geoIDs have matching expression column
all(covar$geoID %in% names(expr)[-1])
```

```
## [1] TRUE
```

```r
# check how many expression probes appear in the linker file
sum(expr$X %in% genes$PROBEID)
```

```
## [1] 500
```

```r
# check whether any probe IDs are repeated in the linker file
sum(duplicated(genes$PROBEID))
```

```
## [1] 0
```

Merge names with expression data and create analysis dataset

```r
# attach gene names to the expression data by matching probe IDs
expr_named <- merge(genes, expr, by.x = "PROBEID", by.y = "X")

# check that many rows remain and for duplicate gene names
nrow(expr_named)
```

```
## [1] 500
```

```r
sum(duplicated(expr_named$GENE))
```

```
## [1] 0
```

```r
# seshape to long format: one row per gene per sample
expr_long <- pivot_longer(expr_named, cols = -c(PROBEID, GENE), names_to = "geoID", values_to = "expres

# join covariates onto each sample by geoID
dat <- left_join(expr_long, covar, by = "geoID")

# confirm size (78,500 rows)
dim(dat)
```

```
## [1] 78500    13
```

Define categorical variables and find candidate genes

```r
# create vitamin D status, standard deficiency cutoff of 20 ng/mL
dat$vitD_status <- factor(ifelse(dat$mat_vitD < 20, "Deficient", "Sufficient"), levels = c("Sufficient"

# create obesity status,standard BMI cutoff of 30 kg/m2
dat$obese <- factor(ifelse(dat$mat_BMI >= 30, "Obese", "Not obese"), levels = c("Not obese", "Obese"))

# check which known preeclampsia-related genes are on 500-probe array
intersect(c("FLT1", "ENG", "LEP", "PGF", "VEGFA", "PAPPA2", "INHBA", "HTRA1"), expr_named$GENE)
```

```
## character(0)
```

Rank genes by pre-eclampsia difference

```r
# for each gene, compare mean expression between preeclampsia and normal pregnancies
gene_rank <- dat %>%
  group_by(GENE) %>%
  summarise(mean_diff = mean(expression[preg_preeclampsia == "Preeclampsia"]) - mean(expression[preg_pr
            p_value = t.test(expression ~ preg_preeclampsia)$p.value) %>%
  arrange(p_value)

# show 10 genes with strongest difference
head(gene_rank, 10)
```

```
## # A tibble: 10 x 3
##    GENE      mean_diff p_value
##    <chr>         <dbl>   <dbl>
##  1 ATG101       0.0636  0.0141
##  2 PCSK7        0.0700  0.0156
##  3 FOXI1        0.0509  0.0163
##  4 YIPF6       -0.125   0.0172
##  5 LACRT        0.0501  0.0202
##  6 SUFU         0.0528  0.0218
##  7 HNF1A-AS1    0.0549  0.0224
##  8 MOSPD1      -0.139   0.0226
##  9 RUNDC3A     -0.261   0.0279
## 10 CYBC1        0.0539  0.0285
```

```r
# show the range of expression values across genes and samples
summary(dat$expression)
```

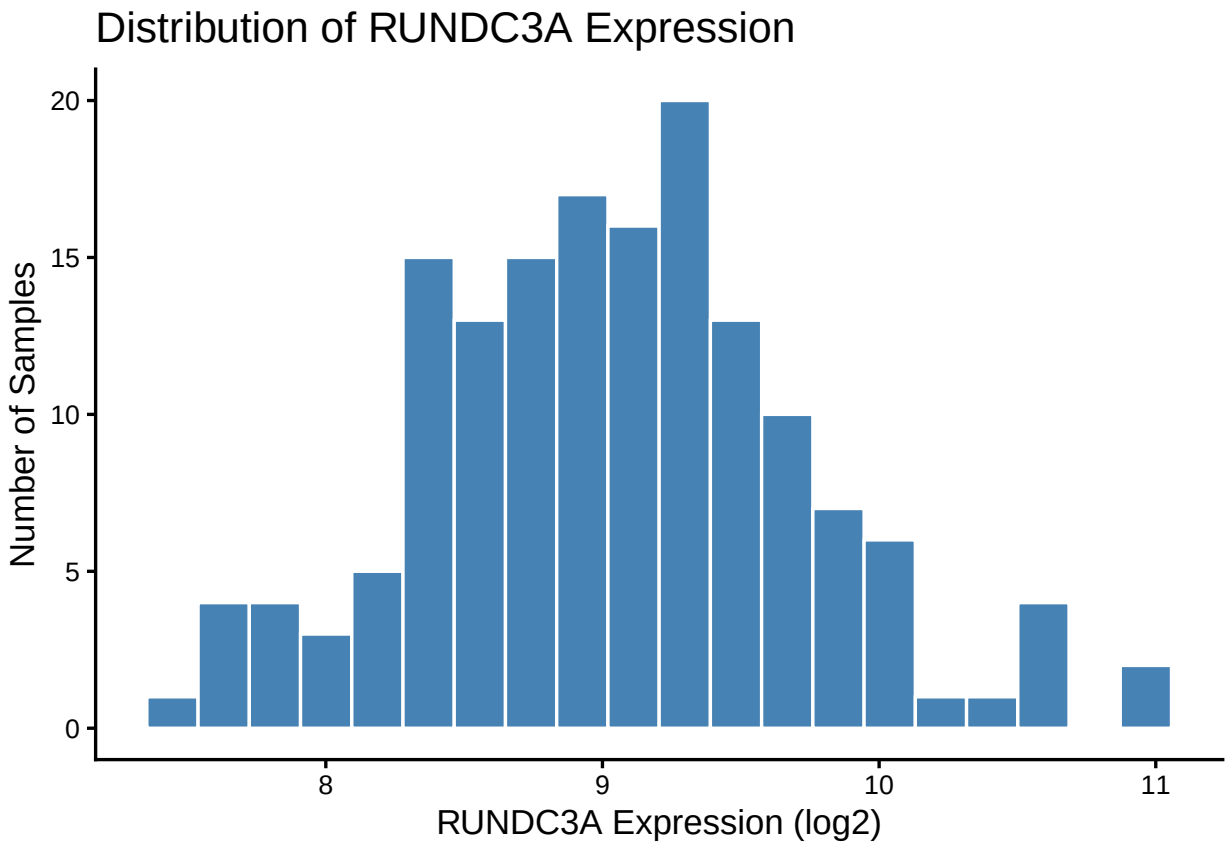
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   3.124   5.725   7.047   7.123   8.290  13.215
```

Gene of interest

```r
# keep only rows for chosen gene
gene_dat <- filter(dat, GENE == "RUNDC3A")
# confirm one row per sample
nrow(gene_dat)
```

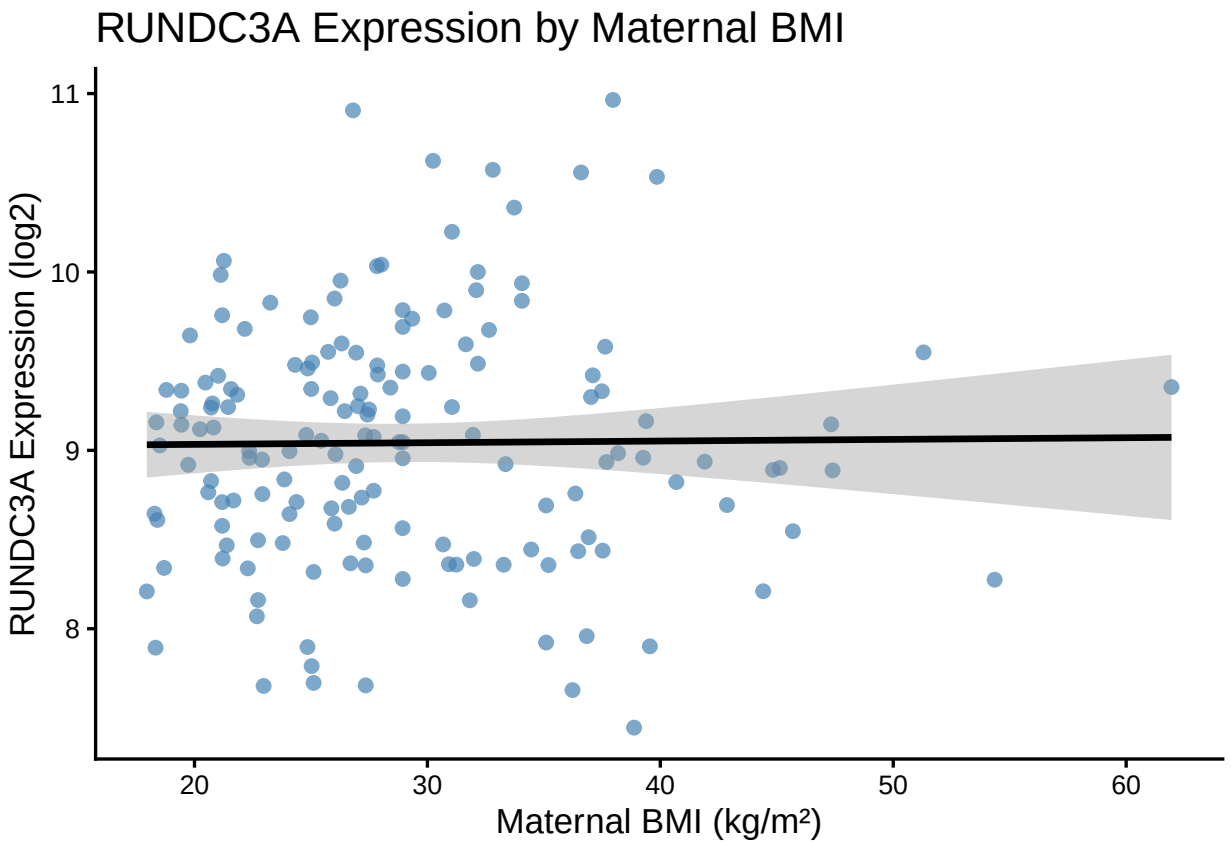
```
## [1] 157
```

```
# plot the distribution of RUNDC3A expression across all samples
p1 <- ggplot(gene_dat, aes(x = expression)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  labs(x = "RUNDC3A Expression (log2)", y = "Number of Samples",
       title = "Distribution of RUNDC3A Expression") +
  theme_classic(base_size = 13)
p1
```



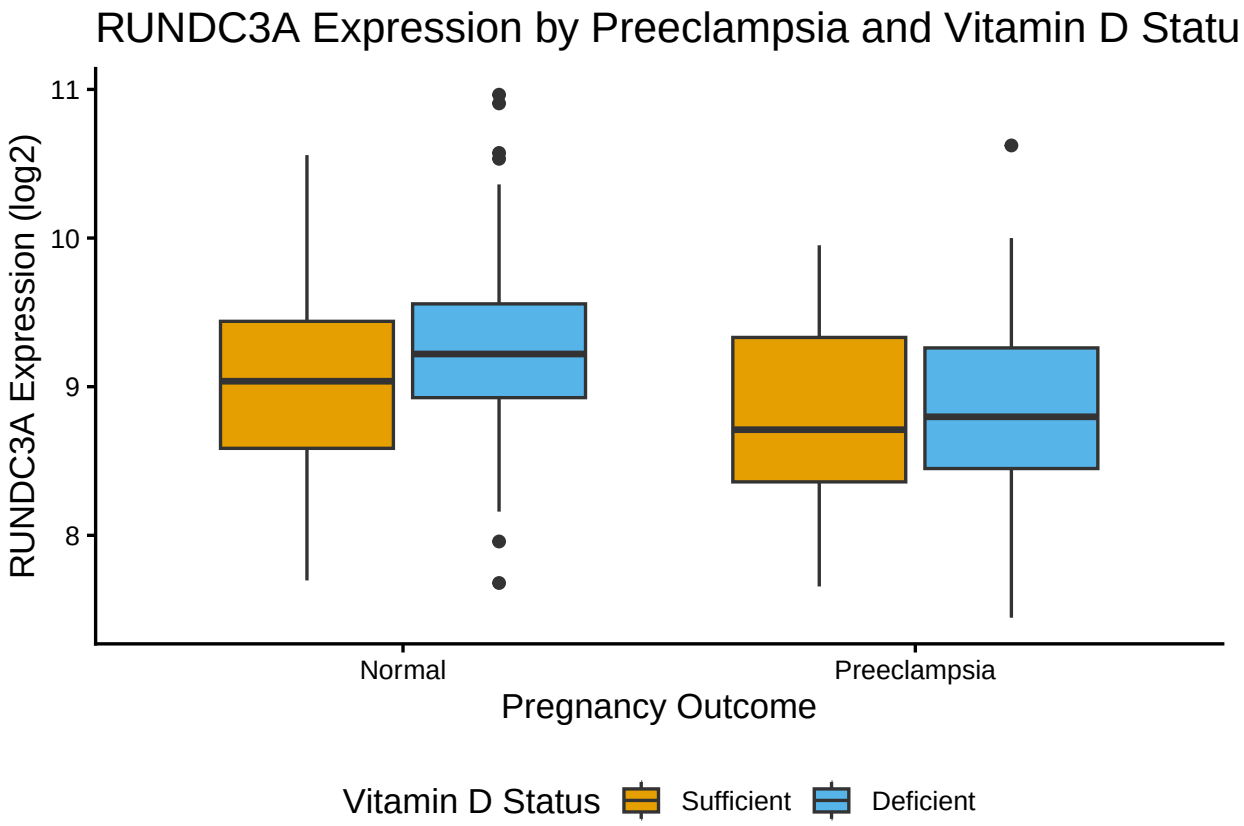
BMI vs Gene Expression

```
# Plot RUNDC3A expression against maternal BMI with a fitted regression line
p2 <- ggplot(gene_dat, aes(x = mat_BMI, y = expression)) +
  geom_point(color = "steelblue", alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black") +
  labs(x = "Maternal BMI (kg/m²)", y = "RUNDC3A Expression (log2)",
       title = "RUNDC3A Expression by Maternal BMI") +
  theme_classic(base_size = 13)
p2
```



Preeclampsia and Vit D status

```
# plot RUNDC3A expression by preeclampsia status, split by vitamin D status
p3 <- ggplot(gene_dat, aes(x = preg_preeclampsia, y = expression, fill = vitD_status)) +
  geom_boxplot() +
  scale_fill_manual(values = c("Sufficient" = "#E69F00", "Deficient" = "#56B4E9")) +
  labs(x = "Pregnancy Outcome", y = "RUNDC3A Expression (log2)",
       fill = "Vitamin D Status", title = "RUNDC3A Expression by Preeclampsia and Vitamin D Status") +
  theme_classic(base_size = 13) +
  theme(legend.position = "bottom")
p3
```



Save figures

```
# Save each plot as png
ggsave("fig1_histogram.png", p1, width = 7, height = 5)
ggsave("fig2_scatter.png", p2, width = 7, height = 5)
ggsave("fig3_boxplot.png", p3, width = 7, height = 5)
```

Submission 2:

Choose genes

```
# RUNDC3A plus the next two from gene_rank
myGenes <- c("RUNDC3A", "MOSPD1", "YIPF6")

# Check that all 3 genes are in data
myGenes %in% dat$GENE
```

```
## [1] TRUE TRUE TRUE
```

Plot labels

```
# labels for variable names
myLabels <- c('mat_BMI' = 'Maternal BMI (kg/m^2)',
              'preg_preeclampsia' = 'Pregnancy Outcome',
              'vitD_status' = 'Vitamin D Status')
myLabels
```

```
##                  mat_BMI       preg_preeclampsia             vitD_status
## "Maternal BMI (kg/m^2)"     "Pregnancy Outcome"      "Vitamin D Status"
```

Plotting function

```
# function to make plots for each gene
plotGenes <- function(data,genelist,contVar,catVars) {
  # loop through all genes
  for(i in genelist) {
    # subset data to this gene
    geneData <- data[data$GENE == i, ]
    # hist of gene expression
    myHist <- ggplot(geneData, aes(x = expression)) +
      geom_histogram(bins = 20, fill = 'steelblue',color = 'white') +
      labs(x = paste0(i, ' Expression (log2)'), y = 'Number of Samples',
           title = paste0(i, 'Expression')) +
      theme_classic(base_size = 16)

    # scatterplot of continuous covariate vs gene expression
    myScatter <- ggplot(geneData, aes(x = .data[[contVar]], y = expression)) +
      geom_point(color = 'steelblue', alpha = 0.7, size = 2) +
      geom_smooth(method = 'lm', formula = y ~ x, color = 'black') +
      labs(x = myLabels[contVar], y = paste0(i, ' Expression(log2)'),
           title = paste0(i, ' vs. BMI')) +
      theme_classic(base_size = 16)

    #boxplot by both categorical covariates
    myBox <- ggplot(geneData, aes(x = .data[[catVars[1]]], y = expression, fill = .data[[catVars[2]]]))
      geom_boxplot(outlier.shape = NA) +
      geom_point(position = position_jitterdodge(jitter.width = 0.2), alpha = 0.5,
size = 1.5) +
      scale_fill_manual(values = c('#E60F00','#56B4E9')) +
      labs(x = myLabels[catVars[1]], y = paste0(i, ' Expression (log2)'),
           fill = myLabels[catVars[2]], title = paste0(i, ' by Outcome and Vitamin D Status')) +
      theme_classic(base_size = 16) +
      theme(legend.position = 'bottom')

# print 3 plots
    print(myHist)
    print(myScatter)
    print(myBox)

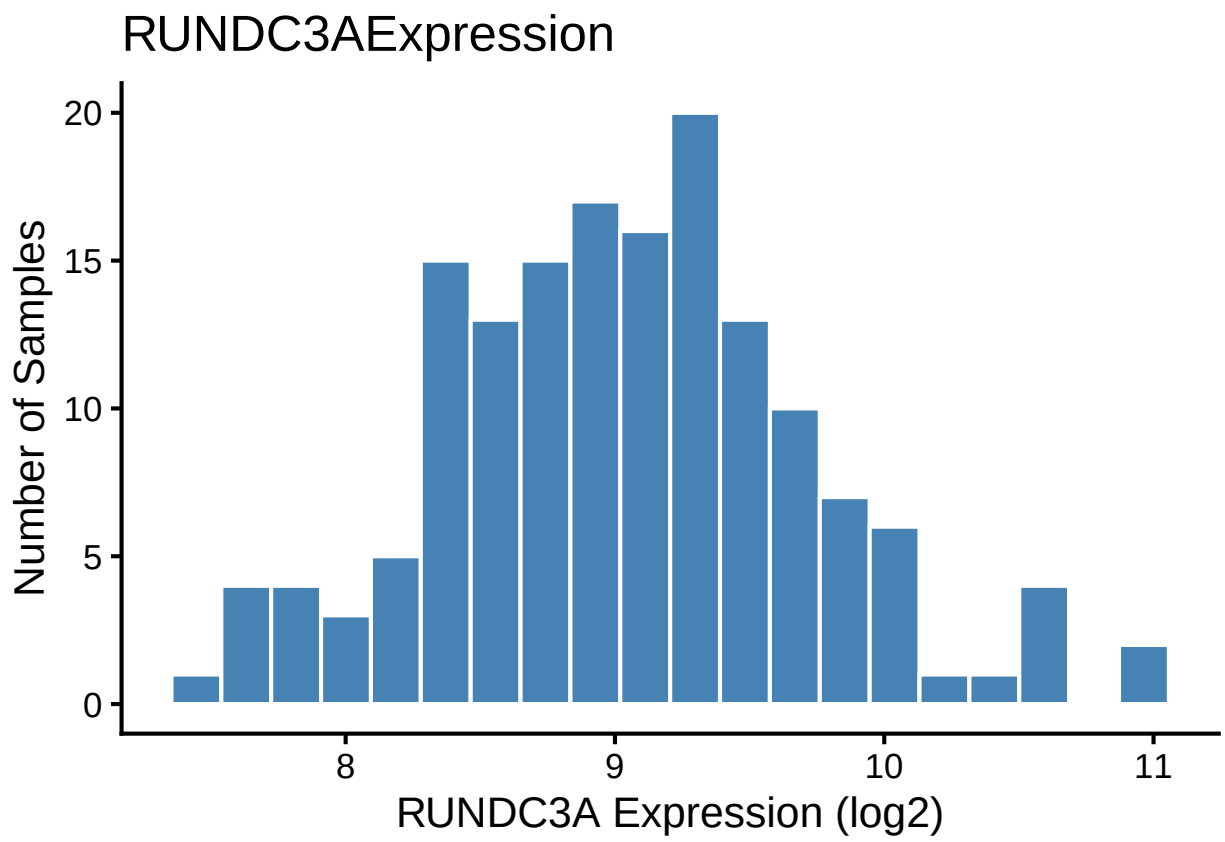
# save 3 plots as png. named by gene
```

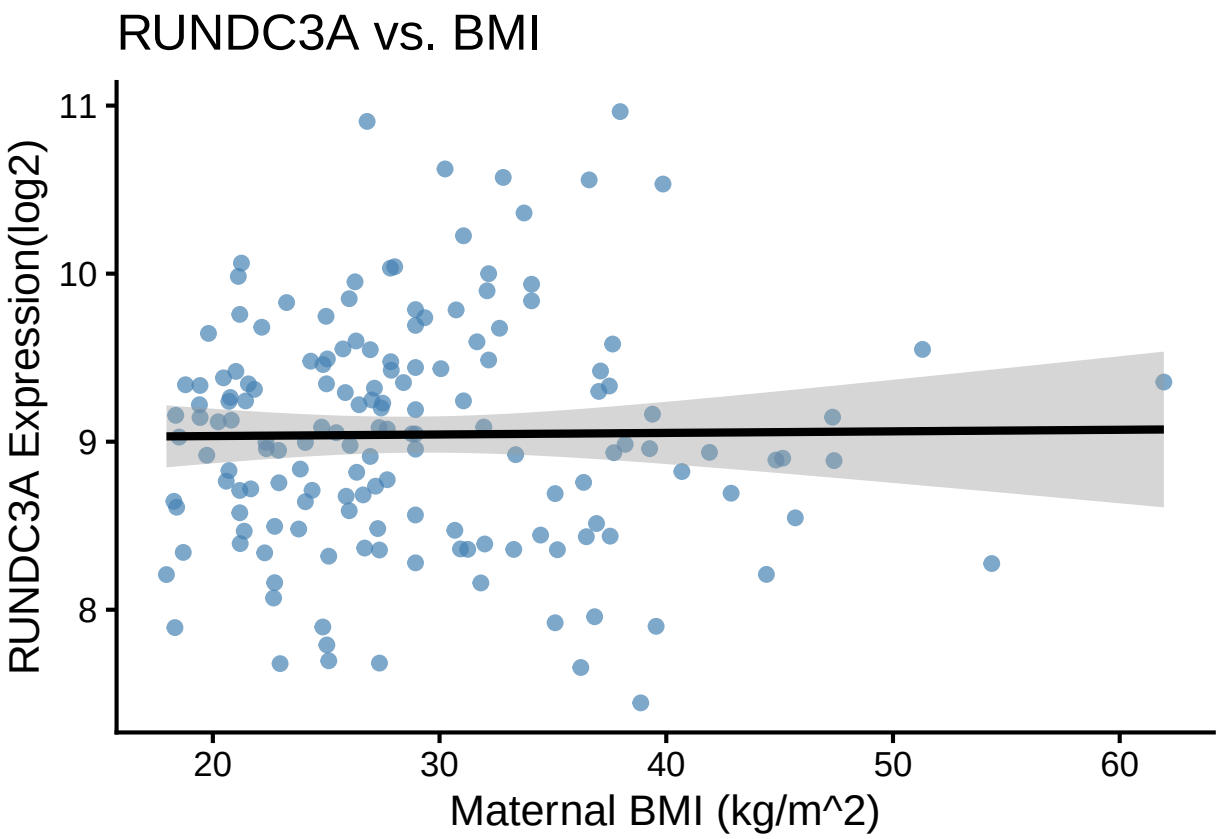
```
ggsave(paste0(i, '_histogram.png'), myHist, width = 7, height = 5)
ggsave(paste0(i, '_scatter.png'), myScatter, width = 7, height = 5)
 ggsave(paste0(i, '_boxplot.png'), myBox, width = 7, height = 5)

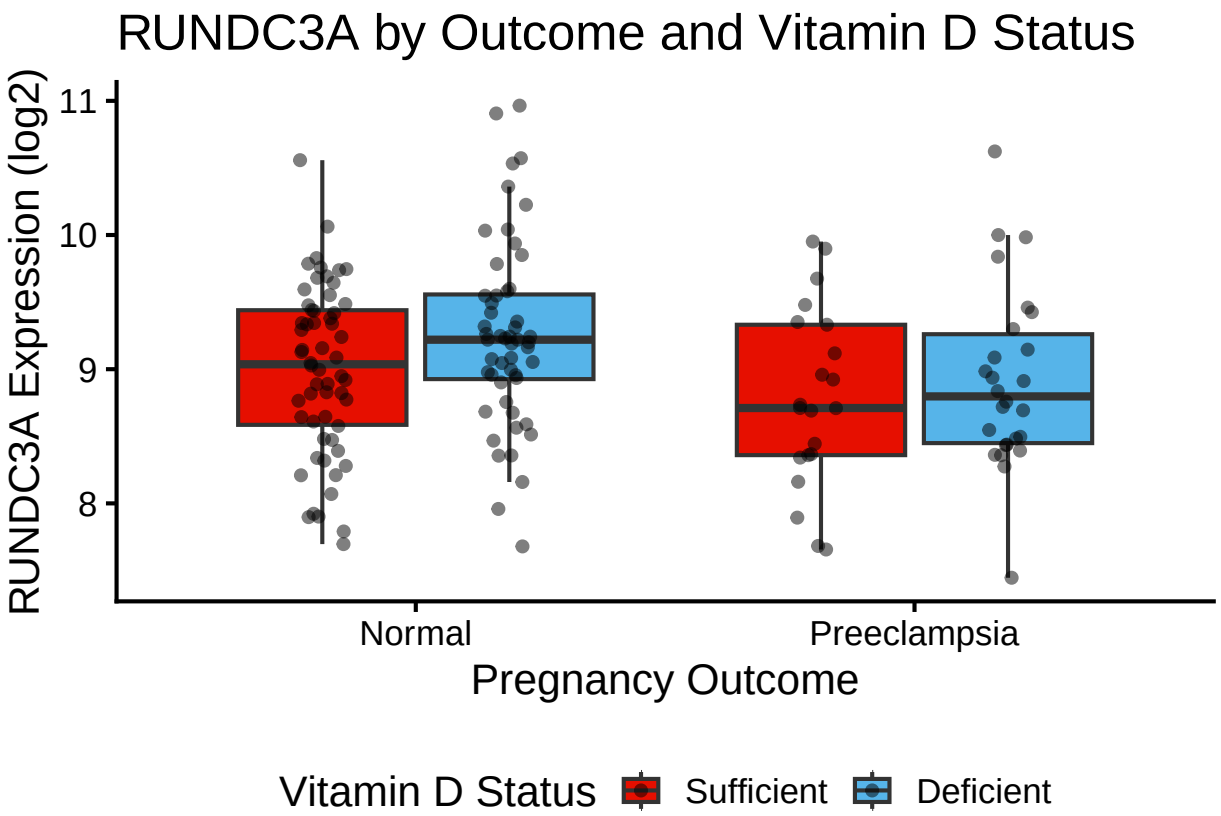
  }
}
```

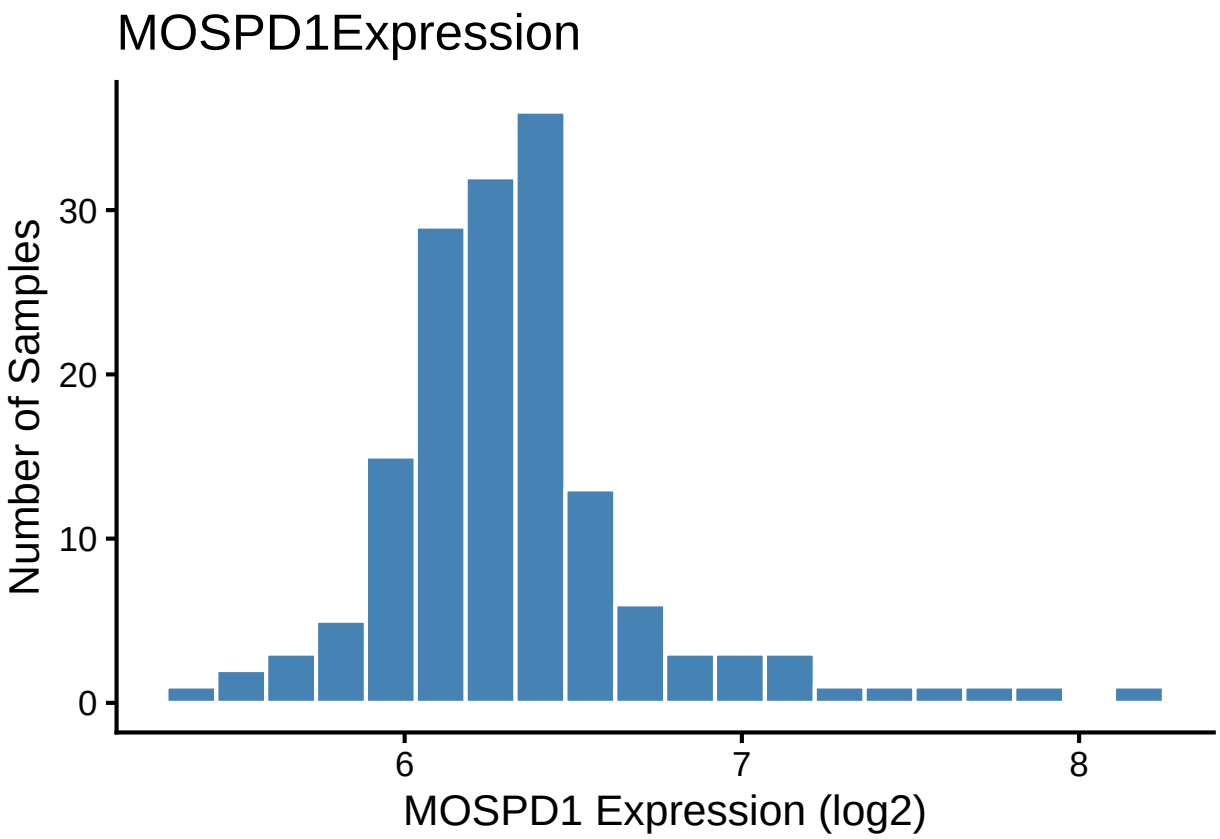
Run the function

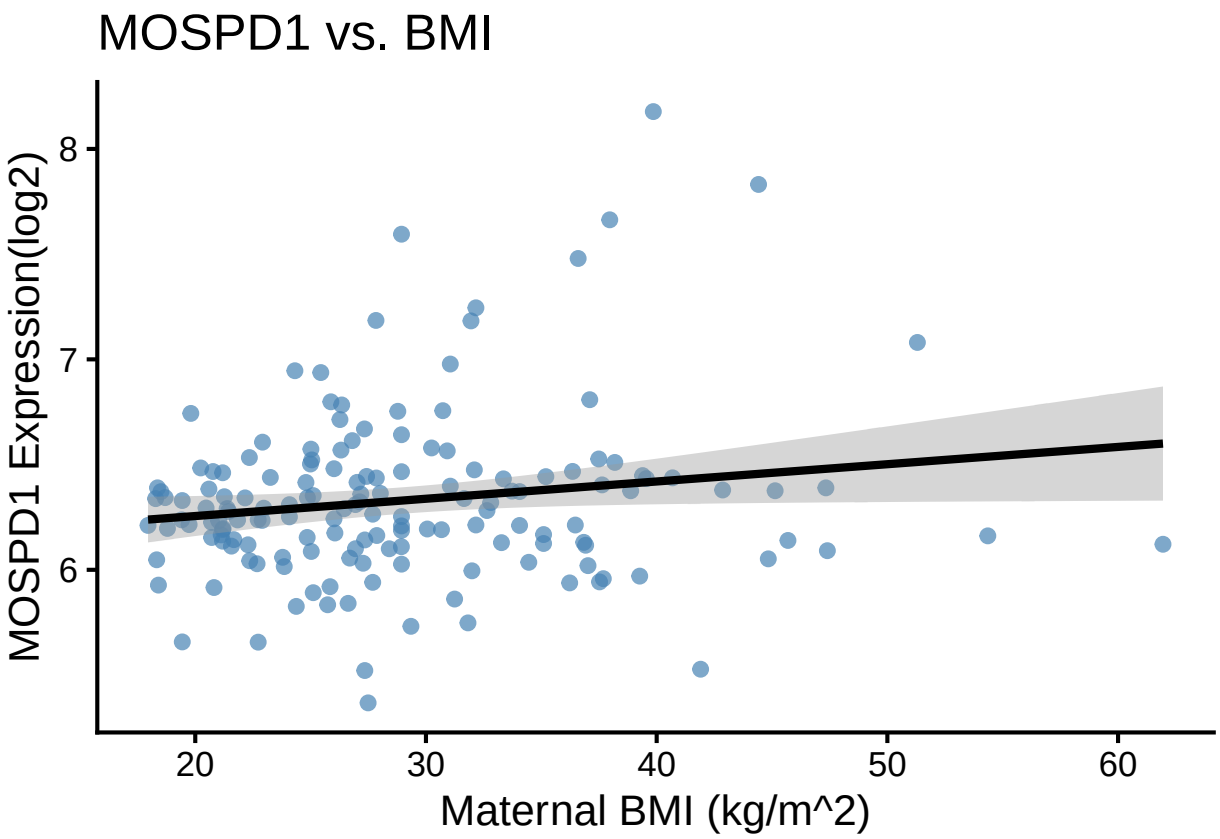
```
# 3 plot types of for each of 3 genes
plotGenes(data = dat,
          genelist = myGenes,
          contVar = 'mat_BMI',
          catVars = c('preg_preeclampsia', 'vitD_status'))
```

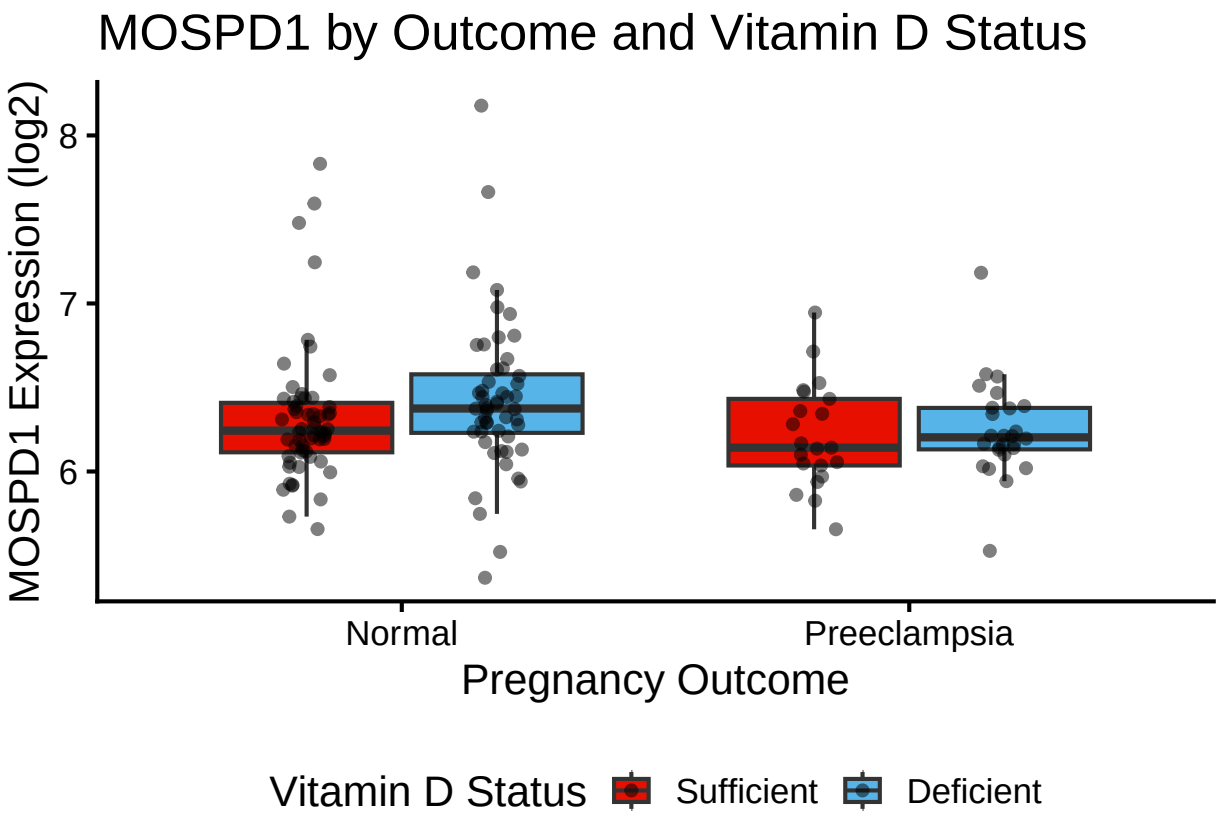


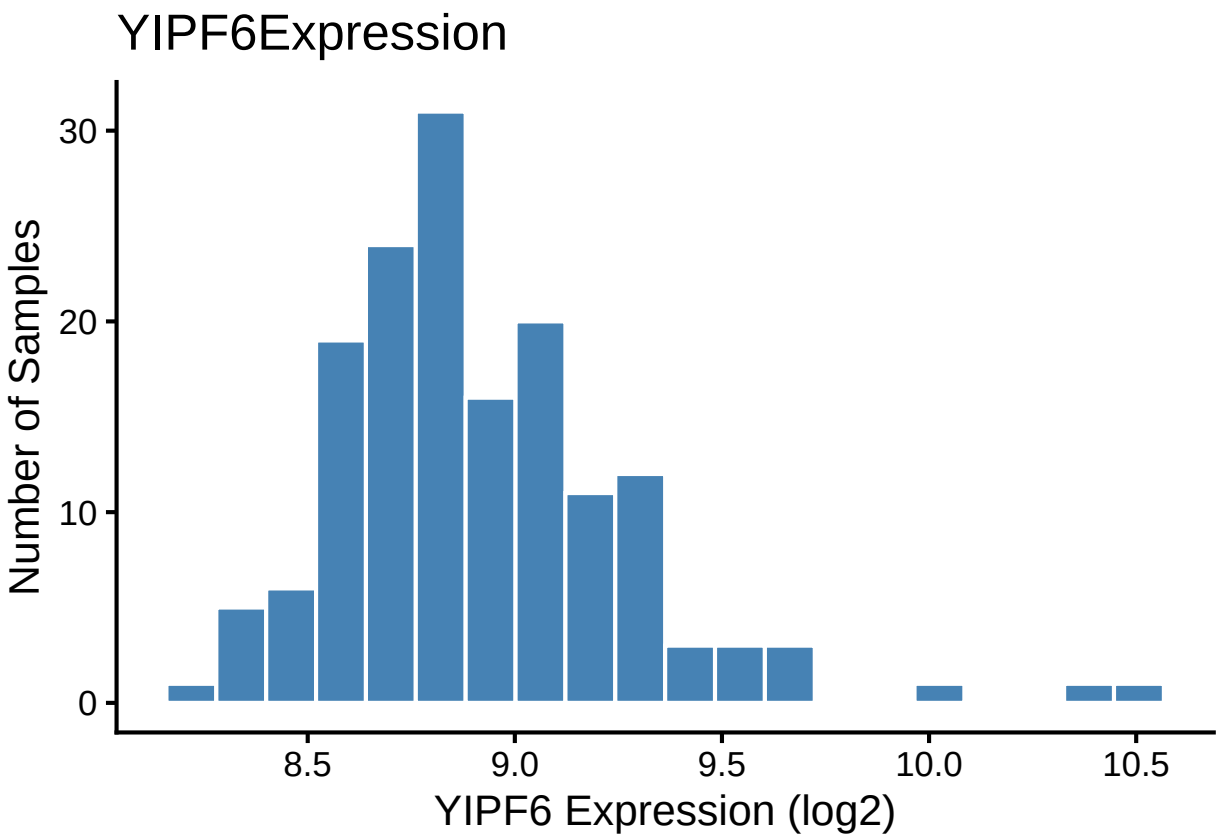

RUNDC3A vs. BMI
RUNDC3A Expression(log2)
11
10
9
8
20
30
40
50
60
Maternal BMI (kg/m^2)

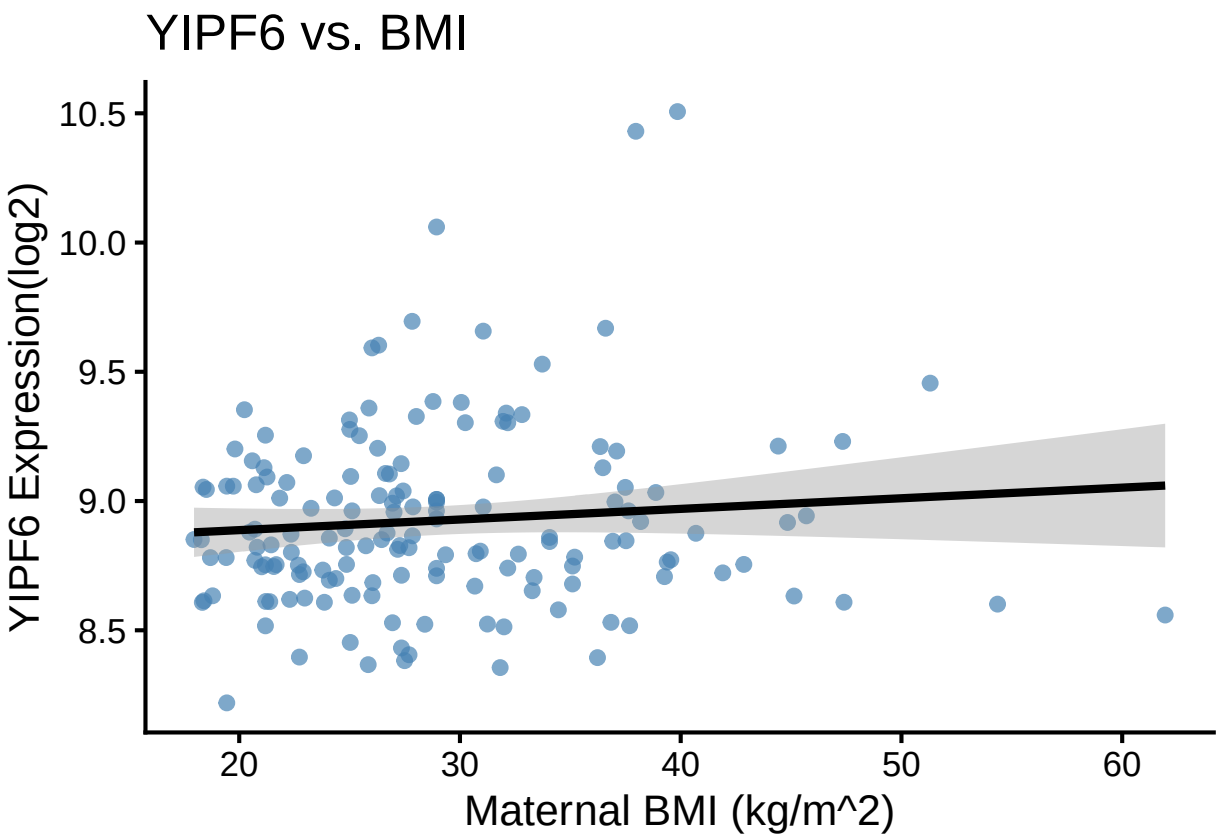

RUNDC3A by Outcome and Vitamin D Status
RUNDC3A Expression (log2)
11
10
9
8
Normal
Preeclampsia
Pregnancy Outcome
Vitamin D Status
Sufficient
Deficient

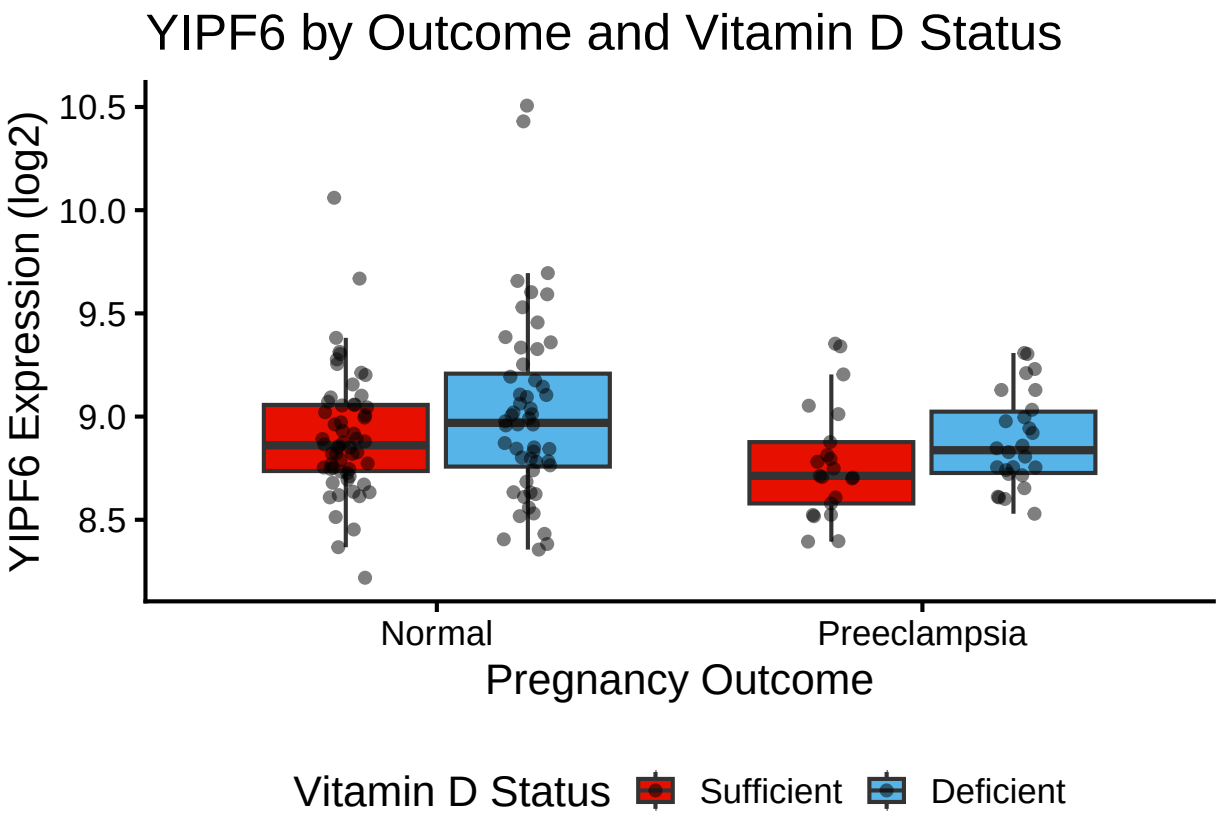

MOSPD1Expression
Number of Samples
30
20
10
0
6
7
8
MOSPD1 Expression (log2)


MOSPD1 vs. BMI
MOSPD1 Expression(log2)
8
7
6
20
30
40
50
60
Maternal BMI (kg/m^2)


MOSPD1 by Outcome and Vitamin D Status
MOSPD1 Expression (log2)
8
7
6
Normal
Preeclampsia
Pregnancy Outcome
Vitamin D Status
Sufficient
Deficient


YIPF6Expression
Number of Samples
30
20
10
0
8.5
9.0
9.5
10.0
10.5
YIPF6 Expression (log2)


YIPF6 vs. BMI
YIPF6 Expression(log2)
10.5
10.0
9.5
9.0
8.5
20
30
40
50
60
Maternal BMI (kg/m^2)


YIPF6 by Outcome and Vitamin D Status
YIPF6 Expression (log2)
10.5
10.0
9.5
9.0
8.5
Normal
Preeclampsia
Pregnancy Outcome
Vitamin D Status
Sufficient
Deficient